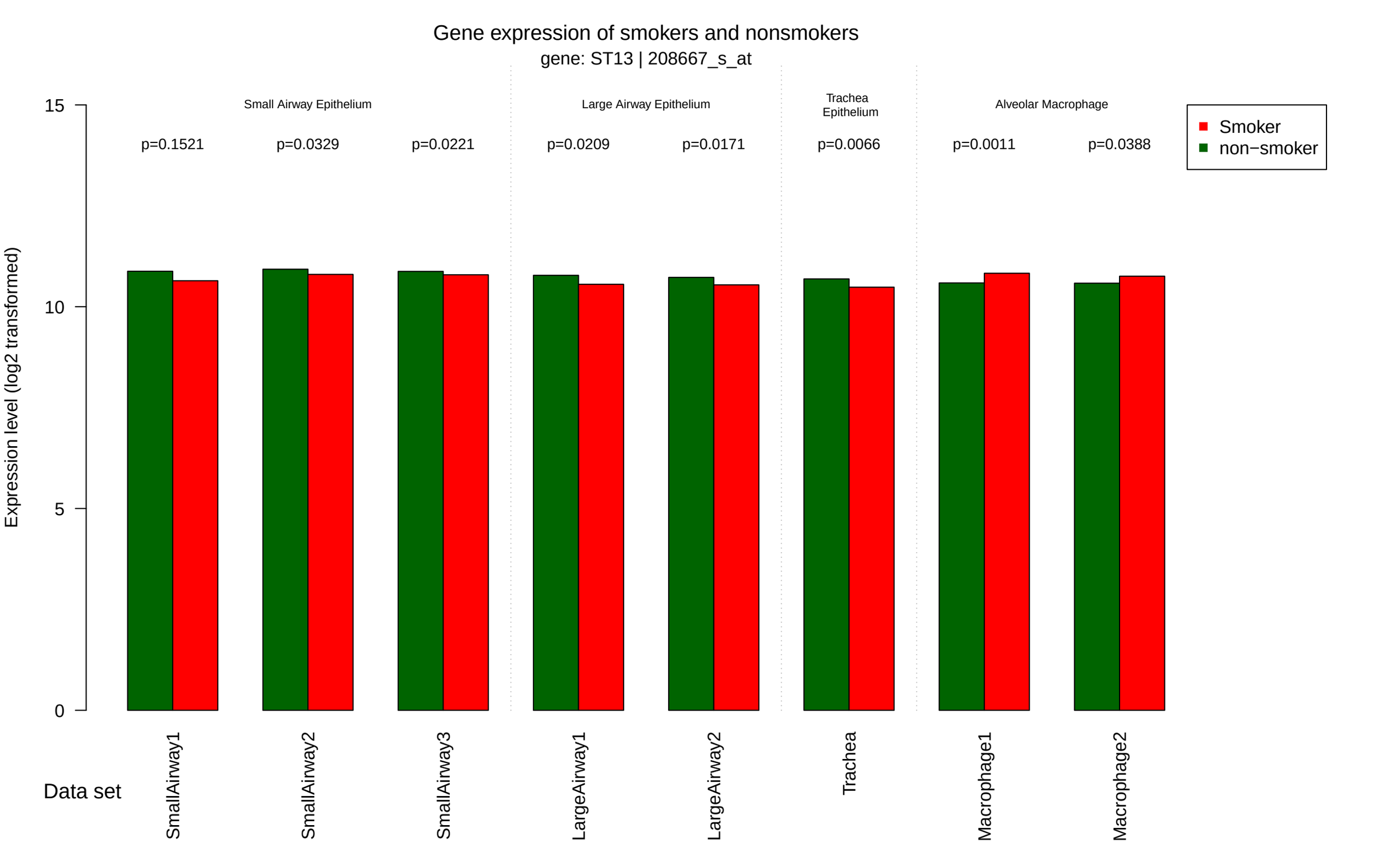
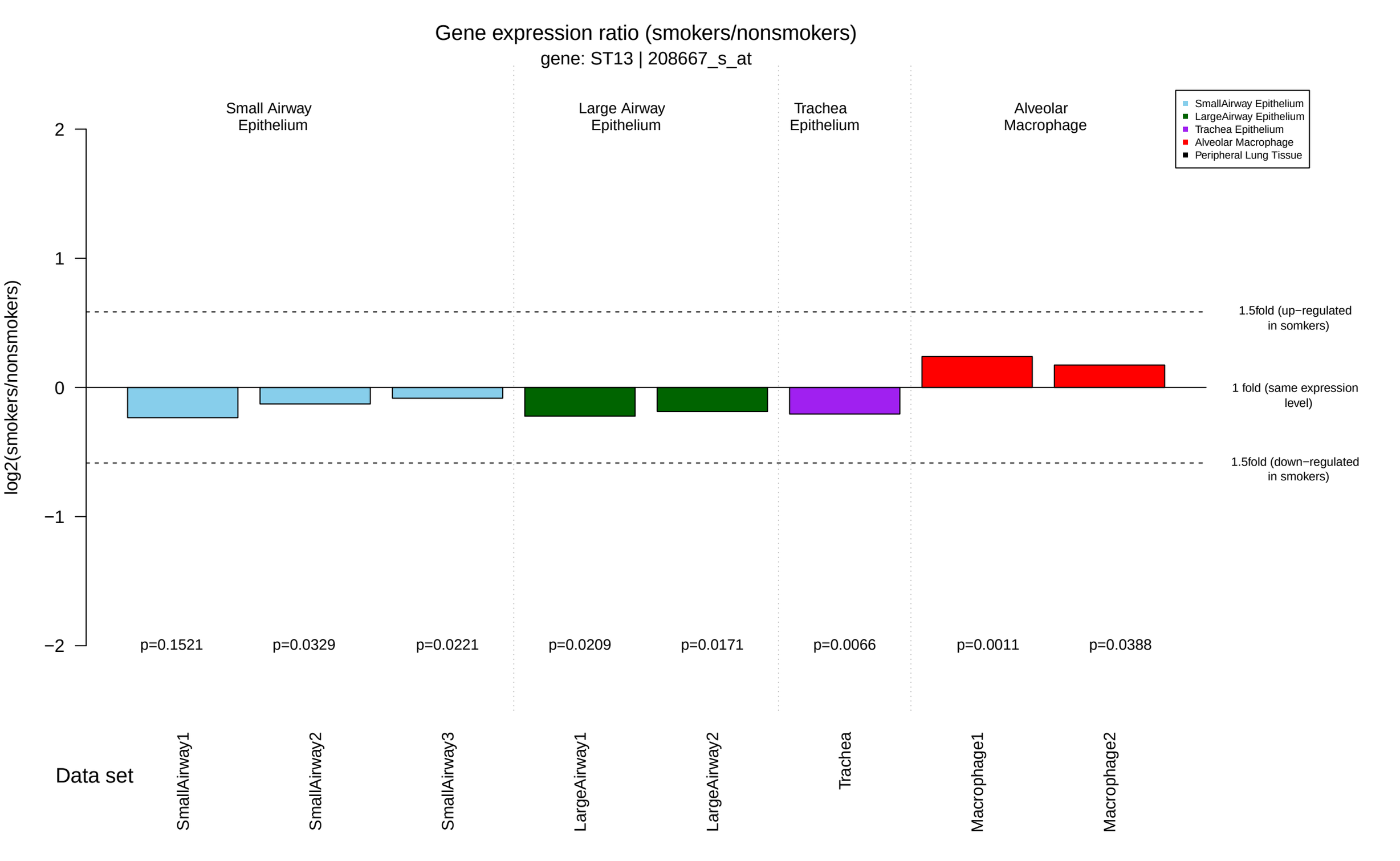
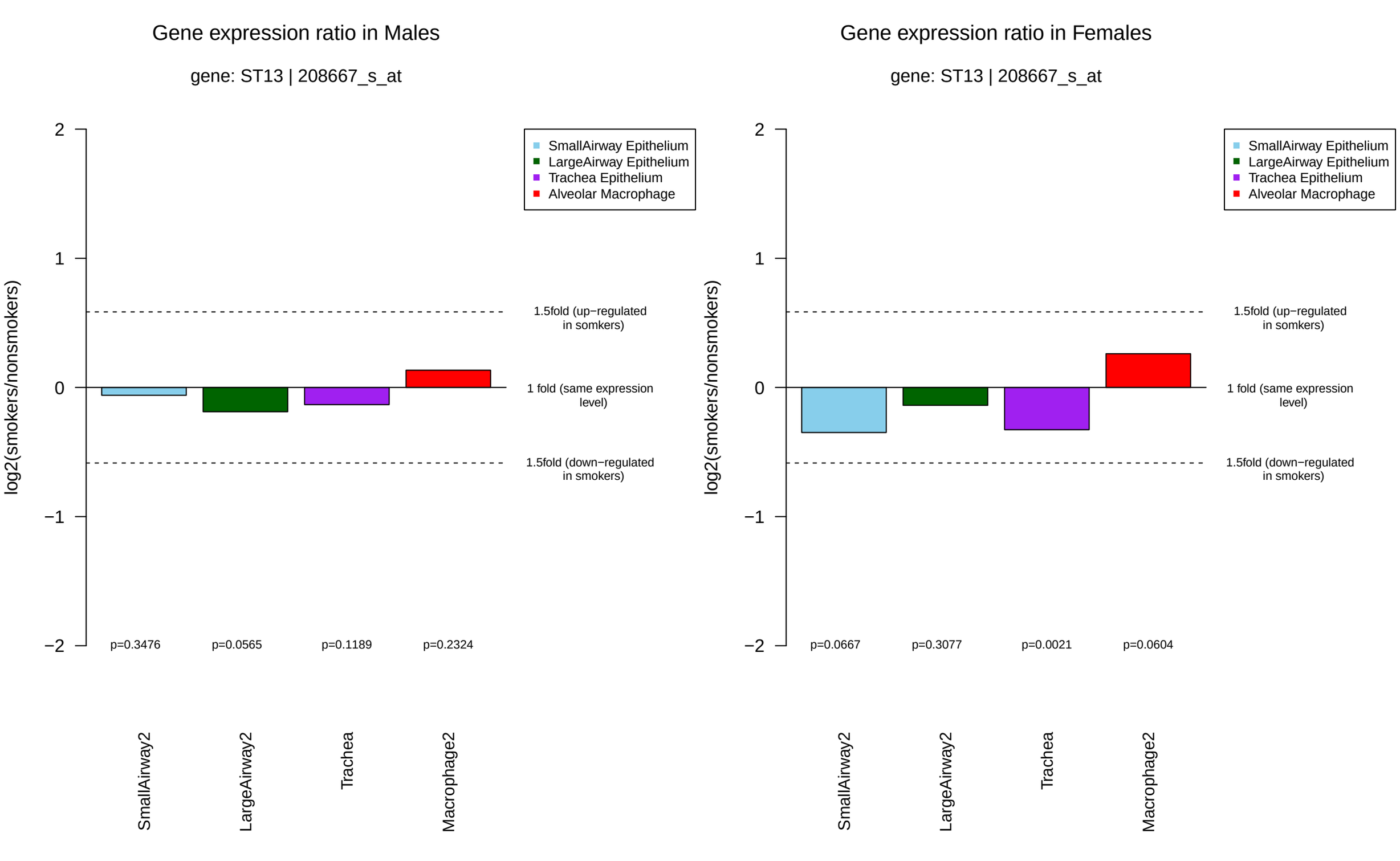
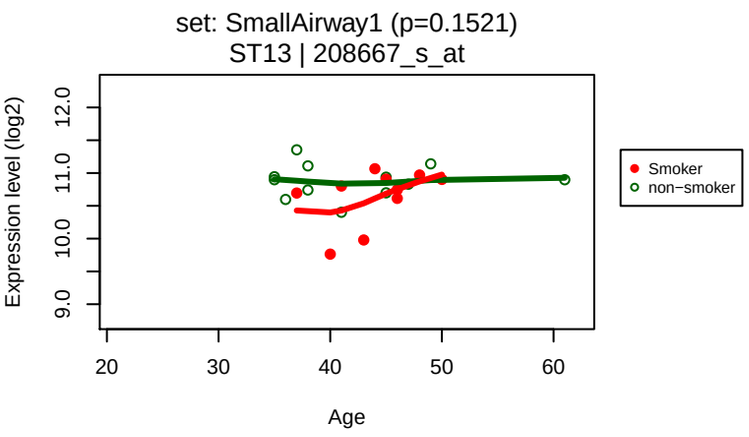
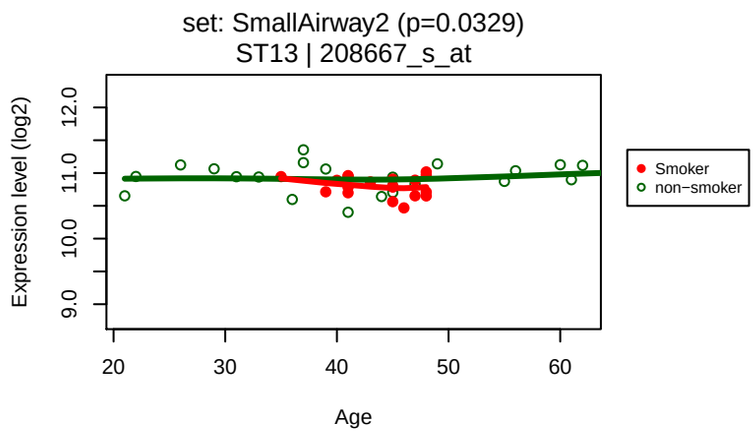
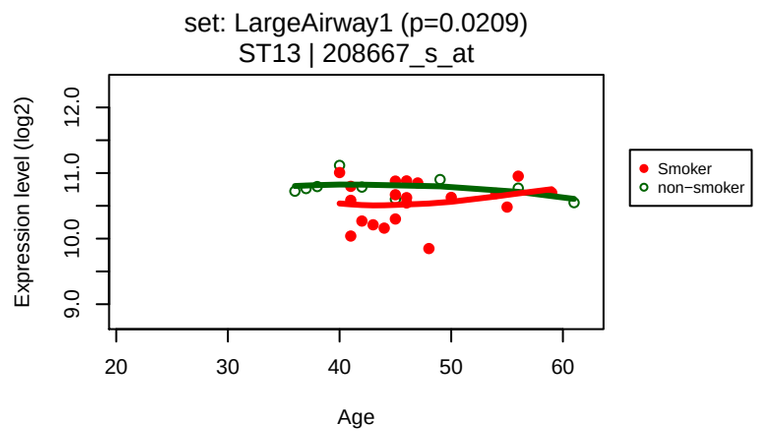

Gene expression of smokers and nonsmokers
gene: ST13 | 208667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1521
p=0.0329
p=0.0221
p=0.0209
p=0.0171
p=0.0066
p=0.0011
p=0.0388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

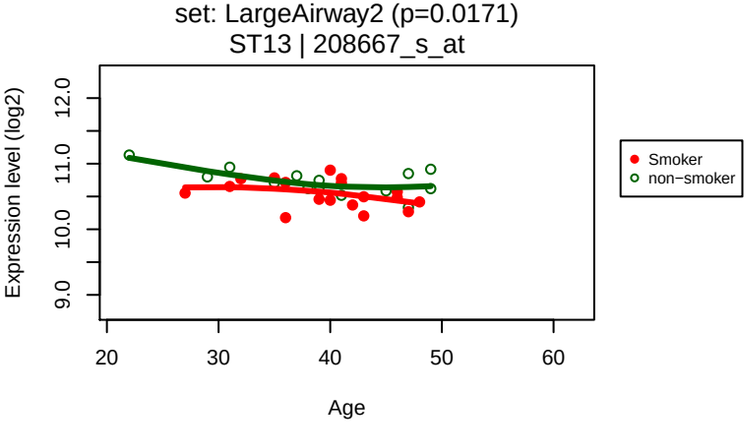

Gene expression ratio (smokers/nonsmokers)
gene: ST13 | 208667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1521
p=0.0329
p=0.0221
p=0.0209
p=0.0171
p=0.0066
p=0.0011
p=0.0388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

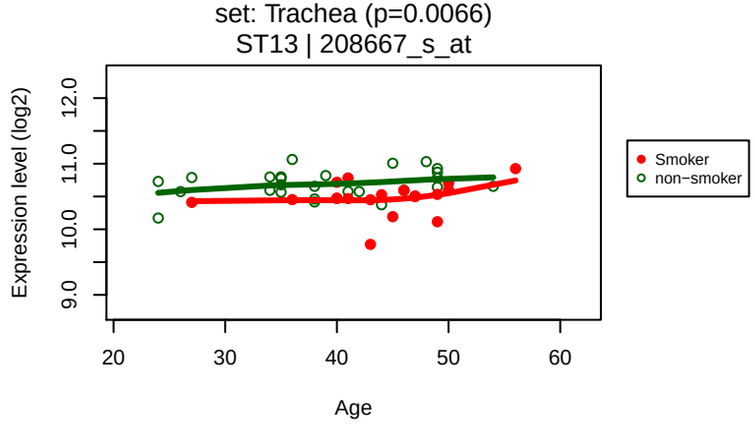

Gene expression ratio in Males
gene: ST13 | 208667_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3476
p=0.0565
p=0.1189
p=0.2324
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST13 | 208667_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0667
p=0.3077
p=0.0021
p=0.0604
SmallAirway2
LargeAirway2
Trachea
Macrophage2

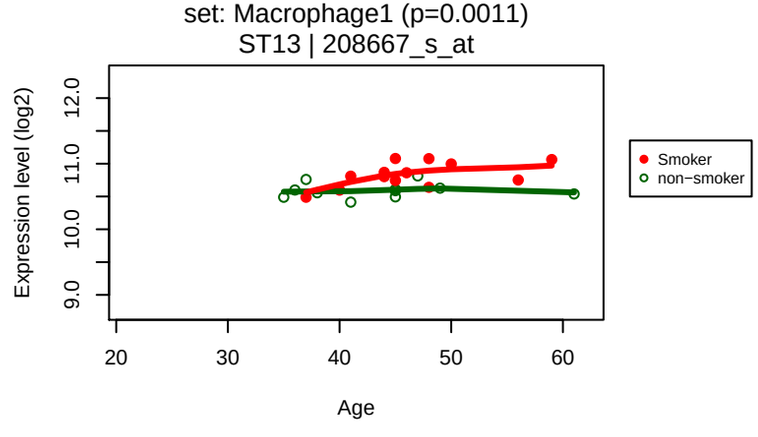

set: SmallAirway1 (p=0.1521)
ST13 | 208667_s_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker

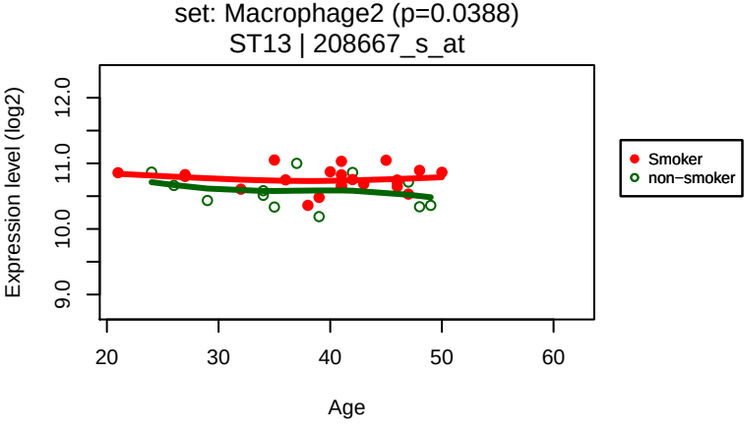

set: SmallAirway2 (p=0.0329)
ST13 | 208667_s_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker

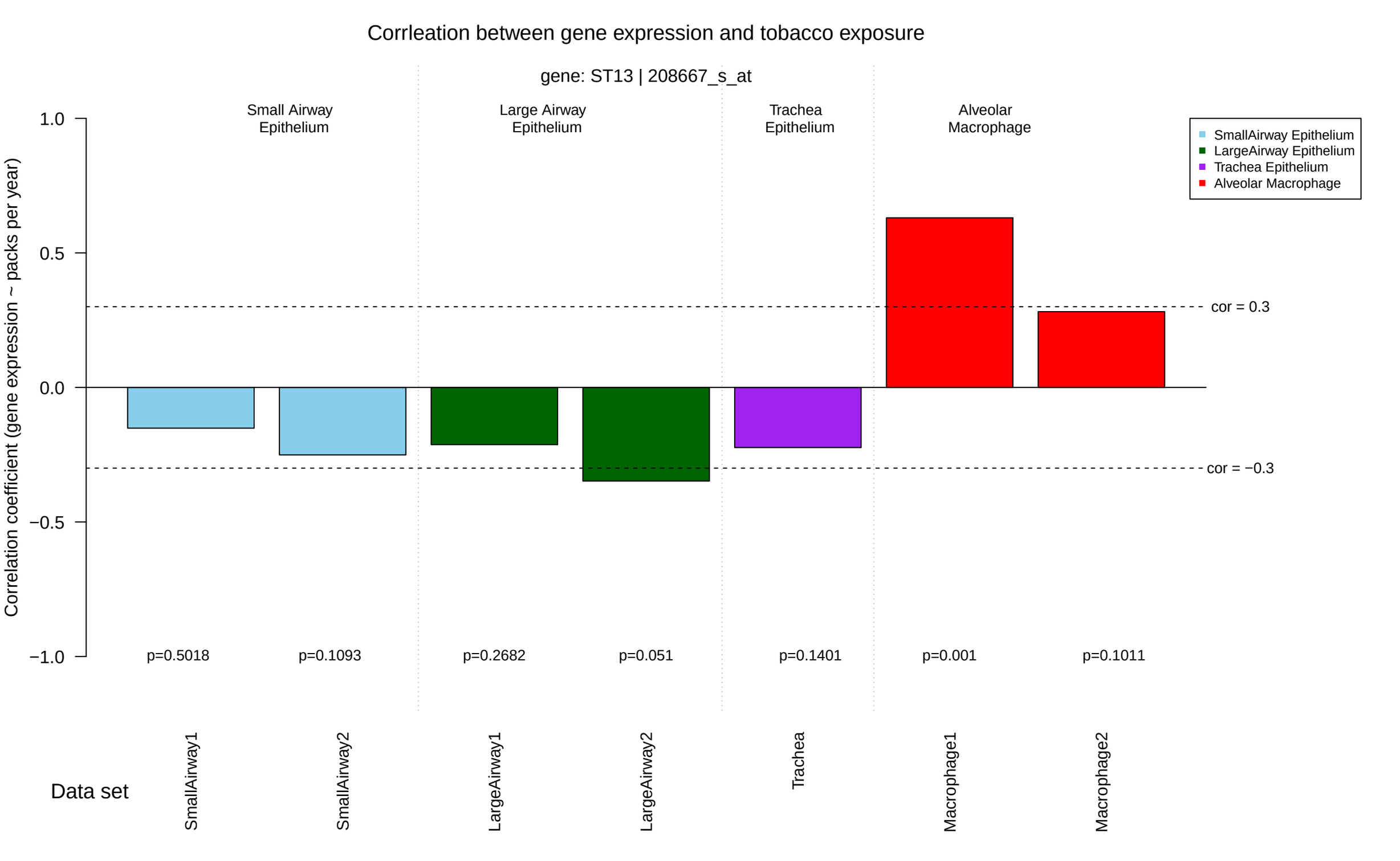

set: LargeAirway1 (p=0.0209)
ST13 | 208667_s_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0171)
ST13 | 208667_s_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0066)
ST13 | 208667_s_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0011)
ST13 | 208667_s_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0388)
ST13 | 208667_s_at
Expression level (log2)
9.0
10.0
11.0
12.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST13 | 208667_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5018
p=0.1093
p=0.2682
p=0.051
p=0.1401
p=0.001
p=0.1011
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2